

How Prolog Works in This Project

A clear, truthful guide for explaining The Logician

ONE-SENTENCE VERSION

Prolog reads facts about the current chess position, applies human-readable rules, and returns prioritized advice - while Stockfish still chooses moves.



What Prolog does in Sentio

Use this slide when someone asks, 'Why is Prolog here?'

1

READS

It receives facts describing the current board: pieces, turn, checks, attacks, defenders, move number, and castling state.

2

REASONS

It applies symbolic chess rules such as: in check, hanging piece, free capture, develop a minor piece, castle, or fight for the center.

3

EXPLAINS

It returns up to six recommendations, sorted by priority, for the Logician panel to display in plain language.

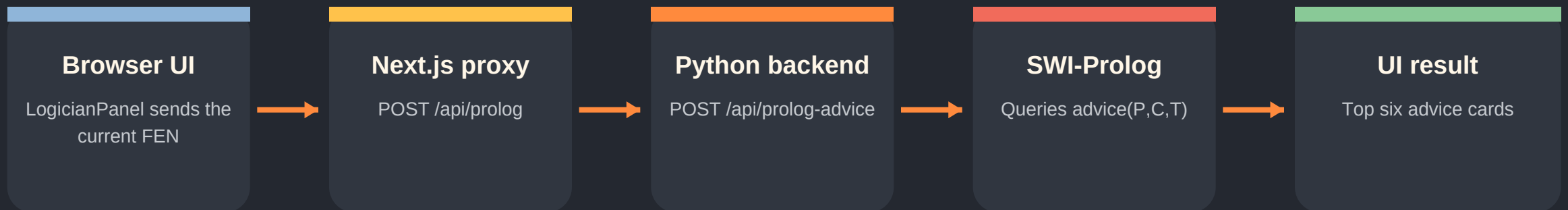
IMPORTANT

Prolog does not calculate the best move, move pieces, or run the voice coach.

It is an explainable reasoning layer beside Stockfish, not a replacement for Stockfish.

Where Prolog sits in the live system

The Logician path is separate from move generation and voice coaching.



STOCKFISH PATH

Stockfish searches legal moves and produces the engine reply. It does not depend on Prolog.

VOICE COACH PATH

Transcription, coach text, and speech use their own API routes. Prolog is not part of the runtime voice-coaching service.

From FEN to prioritized advice

This is the exact request path implemented in the current code.

01 Send FEN

The active Logician tab waits 350 ms, then posts the board FEN to /api/prolog.

02 Validate

The Next.js route validates the body and forwards the FEN to the Python backend.

03 Parse board

python-chess builds a Board object and derives pieces, side to move, check, attacks, and defenders.

04 Refresh facts

The backend retracts old dynamic facts, then asserts facts for this one position.

05 Query rules

SWI-Prolog evaluates `advice(P, C, T)`, with a maximum of 24 raw solutions.

06 Rank and show

Python sorts by priority, returns the top six, and the UI renders category-labelled cards.

What the backend asserts into Prolog

Each request replaces the previous position facts before querying the rules.

FACT	MEANING	EXAMPLE
piece/3	A piece, its color, and type	<code>piece(e4, white, pawn)</code>
turn/1	The side to move	<code>turn(white)</code>
move_number/1	The FEN fullmove number	<code>move_number(7)</code>
castled/1	King is on c-file or g-file after castling	<code>castled(white)</code>
in_check/1	Side to move is currently in check	<code>in_check(white)</code>
attacked_by/2	A square is attacked by one color	<code>attacked_by(f3, black)</code>
defended_by/2	An occupied square has a same-color defender	<code>defended_by(f3, white)</code>

Presenter note: These are temporary, current-position facts - not a permanent game database.

How the rules choose what appears first

Higher priority advice is shown before lower priority advice.

99	SAFETY	Side to move is in check
95	TACTICS	Own piece is attacked and undefended
90	TACTICS	Enemy piece is attacked and undefended
85	OPENING	King should castle before move 12
80	OPENING	Minor piece remains on its home square
75	OPENING	Queen moved out very early
70	STRATEGY	No friendly piece occupies the four center squares
60	ENDGAME	King remains on the back rank from move 25

Why priorities matter: Immediate danger outranks general opening or strategic advice.

WORKED EXAMPLE

How one hanging knight becomes advice

A simple rule trace you can explain without showing the entire codebase.

INPUT FACTS

```
turn(white).  
piece(f3, white, knight).  
attacked_by(f3, black).  
% no defended_by(f3, white)
```

The absence of a defender is meaningful.

DERIVED RULE

```
hanging_mine(  
    f3, knight  
).
```

The rule checks:

- It is my piece
- It is attacked
- It is not defended
- It is not a king

OUTPUT

PRIORITY 95 - TACTICS

**Your knight on f3 is
attacked and undefended
- defend it or move it.**

Stockfish vs Prolog vs the voice coach

Three AI approaches, each responsible for a different job.

STOCKFISH

Search AI

- Calculates strong moves
- Uses numeric evaluation and search
- Controls the engine reply
- Best at: move quality

PROLOG

Symbolic AI

- Applies hand-written chess rules
- Returns traceable recommendations
- Feeds The Logician panel
- Best at: explainability

VOICE COACH

Language AI

- Answers player questions
- Produces natural-language guidance
- Uses separate transcribe/coach/TTS routes
- Best at: conversation

Best presentation phrase: Stockfish moves, Prolog explains, and the voice coach converses.

Limitations and fallback behavior

These points protect you from overclaiming during questions.

SAFE TO SAY

- Prolog explains selected chess principles from the current board state.
- The advice is deterministic and easy to trace back to a rule.
- The Logician is optional; the rest of Sentio still works if it is unavailable.
- The rule set can be extended with more tactical and strategic concepts.

DO NOT CLAIM

- Prolog calculates the best chess move.
- Prolog controls piece movement or game rules.
- Prolog powers every AI feature in Sentio.
- Prolog is part of the runtime voice coach.

Fallback: Missing SWI-Prolog, pyswip, or python-chess returns available=false; the UI shows a setup hint.

A 60-second talk track + likely questions

Practice the left side once, then use the answers on the right for Q&A.

SAY THIS

Sentio uses Prolog for The Logician, its explainable chess-advice panel. Whenever the board changes, the panel sends the current FEN through our Next.js API to the Python backend. The backend converts that position into facts - such as which pieces exist, whose turn it is, whether the king is in check, and which pieces are attacked or defended. Prolog applies prioritized rules to those facts and returns clear advice. For example, check has priority 99, while an undefended piece has priority 95. The important distinction is that Stockfish still calculates the move. Prolog explains chess principles in a deterministic, traceable way.

LIKELY Q&A

Why Prolog?

Its rule-based logic is transparent and easy to explain.

Does it choose moves?

No. Stockfish performs the move search.

How is the board connected?

FEN becomes temporary Prolog facts for each request.

What if Prolog is missing?

The Logician shows a setup hint; the rest of Sentio continues.

How can it improve?

Add rules for forks, pins, king safety, pawn structure, and richer context.